Algebra II with Trigonometry Chapter 6.3

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file: 4531393_alg-2trig-h-chp-6-3-note.pdf

Here are 10 mental math questions based on the core concepts from the PDF on trigonometric graphs:

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1. Periodicity

- What is the period of the function $\sin t$?

2. Sine and Cosine at Zero

- What is the value of $\cos 0$ and $\sin 0$?

3. Amplitude Identification

- If the function is $y = 5\sin x$, what is its amplitude?

4. Horizontal Stretch/Shrink

- For $y = \sin(3x)$, is the graph stretched or shrunk horizontally compared to $y = \sin x$? By what factor?

5. Period Calculation

- What is the period of $y = \cos(2x)$?

6. Vertical Shift

- If a sine curve is given by $y = 2 + \sin x$, what is the vertical shift?

7. Phase Shift

- For the function $y = \sin(x - \frac{\pi}{4})$, what is the direction of the phase shift and by how much?

8. Interval for One Period

- For $y = \sin(4x)$, what is an appropriate interval to graph one complete period?

9. Maximum and Minimum Values

- What are the maximum and minimum values of $y = -3\cos x$?

10. Frequency from Period

- If the period of a function is 0.01 seconds, what is its frequency?

These can all be answered without paper, focusing on the most fundamental properties and transformations of trigonometric functions.